Natural Language Project

*Help the doctor*

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# 1 Goals

**Simulate a participation in a shared task (max score: 4)**

As you have seen in classes, shared tasks are competitions in which participants test their systems in specific tasks, using the same test sets. Training sets are given in advance, and, on a certain predefined date, a test set is released. Then, participants have a short period of time to return the output of their systems, which are evaluated and straightforwardly compared with one another, resulting in a final ranking where the state-of-the-art system is acknowledged. We will simulate such an evaluation forum, although the test set (without the expected output) will be released along with the train set.

**Develop your critical reasoning, your communication skills, and your creativity (max score: 16)**

Write a short paper (max 3 pages) in which you describe the models you created, present the obtained results, discuss the latter, and highlight limitations associated with the given data and your system. You should convince us that you have looked at the data and at the returned outputs (not just at the evaluation scores).

# 2 Tasks

This project is about associating a clinical text to a medical specialty. Your task is:

a) **to build models** that map each clinical text (and other given elements related with the clinical text) to its corresponding medical specialty. There are 12 specialties in total (as Surgery, Orthopedic, Radiology, Gastroenterology, Neurology);

b) **write a up to 3 pages short paper** describing your work.

# 3 Your Model

You are given a training set (train.csv) of medical situations, where each line has the following format:

**medical_specialty description sample_name transcription keywords**

- medical_specialty is the label of one of the 12 specialties
- description is a short description of the clinical case or procedure
- sample_name is the name of the procedure or consultation type
- transcription is the full clinical note (free text)
- keywords is a list of keywords associated with the case

For the automatic evaluation of your project, you should run your **best model** on the given test set (test_no_labels.csv). Notice that it has no labels, that is, no "medical_specialty". Anyway, it should never be used for training. The system should return an output file (**that you should name results.txt**), each line with:

**predicted_medical_specialty**

Notice that the line number in which a medical situation appears in the test file (test_no_labels.csv) should be the same line number of the corresponding medical_specialty in results.txt (the automatic evaluation depends on this).

# 4 Evaluation

**Automatic Evaluation (max 4 points):**

If you beat a weak baseline (Multinomial Naive Bayes with TF-IDF features extracted from the "description" field, with lowercasing and stemming) by obtaining an accuracy above 49%, you will receive 2.0 points. An additional 2.0 points will be awarded if, additionally, no per-class F1 score falls below 25% – ensuring that your model predicts all categories and does not collapse to a single dominant class.

**Short paper Evaluation (16 points):**

The short paper should be named NUM.pdf, where NUM is your group number. It must not exceed 3 pages[1], must be written in Portuguese or English, and must follow the

[1] If the paper exceeds 3 pages, only the first three will be evaluated — even if the first page is a cover page with your names and group number.

provided template organization including a data augmentation process (mandatory). **The expected content and score breakdown for each section are detailed in the template.**

Notice that 3 points will be discounted if any instruction is not followed.

# 5 Submission

On October 16th, 2026, you should deliver, until 11:59 PM (23h59), via Fenix, a zip file (NOT a rar) containing the project, named after the group number NUM (ex: 3.zip). The zip file should contain:

- the file NUM.pdf with the short paper;
- the project code (not the models, just the code) and eventual datasets gathered to the project;
- the file named results.txt with the results from the given test set, that is, a list of the labels returned by your best model when it was applied to the given test set (test_no_labels.csv). No headers.

# 6 Details

- This project should be done **in groups up to 4 students** (cooperation). If you are looking for a colleague to create a group, please check the sheet we will make available at Fenix.
- Although we will probably not run your code, you should be prepared to do it. If you are unable to replicate your own work, you will have zero in the automatic component.
- As usual, questions should be sent to meic-ln@disciplinas.tecnico.ulisboa.pt (subject: Project).
- We might release FAQs about the project – again, Fenix (Section Projects). Stay tuned!
- All projects may be subject to an oral discussion. If students are unable to demonstrate a clear understanding of the work they have submitted, the final grade may be adjusted to reflect the demonstrated level of knowledge.

# 7 Comments/tips

- Check the template. There is important information there.
- This is not a BSc project; this is a MSc project: there is a clearly identified problem that you need to solve in the best possible way, but we do not tell you how to do it.
- Remember what you have learned during the class about data and evaluation: try to do a systematic work.
- Pre-processing applied to the training set should also be applied to the test set.
- Attention to blindly removing stop words.
- There is no 100% accuracy (this is a research problem).
- The dataset is adapted from a "real" dataset from Kaggle[2]. Datasets in NL have errors and are usually imbalanced. You will also probably find many labels that you don't agree with. You are probably right, but the datasets will not be changed. Discuss these situations in your paper.
- Look at the input and output!!!!!!!!!!!!!!!!!! (not just to numbers)

Thank you!
We really hope you **enjoy** the project and have a **good learning experience** with it! ♥

[2] https://www.kaggle.com/